

NORDIC CONFERENCE ON FUTURE HEALTH 2026 · 14-16 SEPTEMBER

Nordic Biobank × NVIDIA Federated Learning Hackathon for Biomedical Applications

Cogni Warriors

Predicting cognitive progression from routine phenotypes and imaging —
across institutions, without moving a single record.

Multimodal late fusion · random-effects loss · NVIDIA FLARE 2.9 FedAvg · [open on GitHub](#)

THE QUESTION

Can low-burden measures predict executive function at every visit?

The cohort

Penn LEAD (OpenNeuro ds007116) — 132 adolescents, 225 imaging sessions, up to 3 visits over ~1.5 years.

Three diagnostic groups. The modelling unit is the **session**, not the subject.

Inputs: 91 phenotype features plus an 80-dim structural-MRI embedding.

The target

An executive-function composite: the mean of 5 z-scored Penn CNB tasks — N-back, PCET, AIM, CPT, Trail Making B.

Valid for **217 of 225 sessions** (at least 2 of 5 tasks passing QC).

Continuous and transdiagnostic — diagnosis is deliberately not the outcome.

Repeated visits → correlated observations

Data siloed across institutions

Centres instrument differently

No leakage: max |Pearson r| between any phenotype feature and the composite, or any of its 5 components, is 0.58.

PREPROCESSING

From raw scans and 42 phenotype files to one row per session

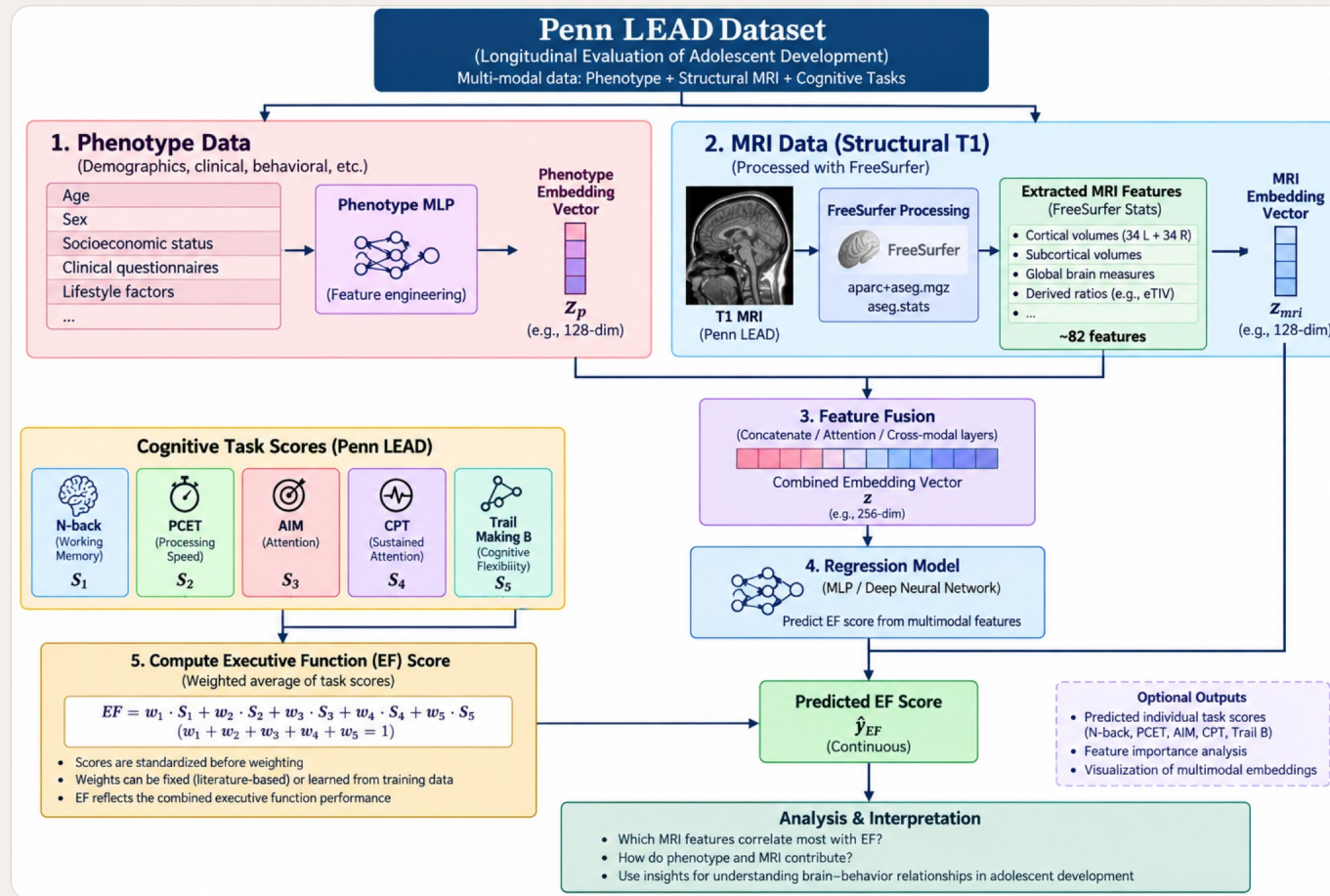
Penn LEAD — Data Preprocessing Pipeline

From raw MRI scans to modeling-ready data with target



THE PIPELINE

Phenotype and imaging in, a continuous EF score out



Two branches, one score

Phenotype and FreeSurfer-derived MRI features are embedded separately, then fused.

EF is built, not borrowed

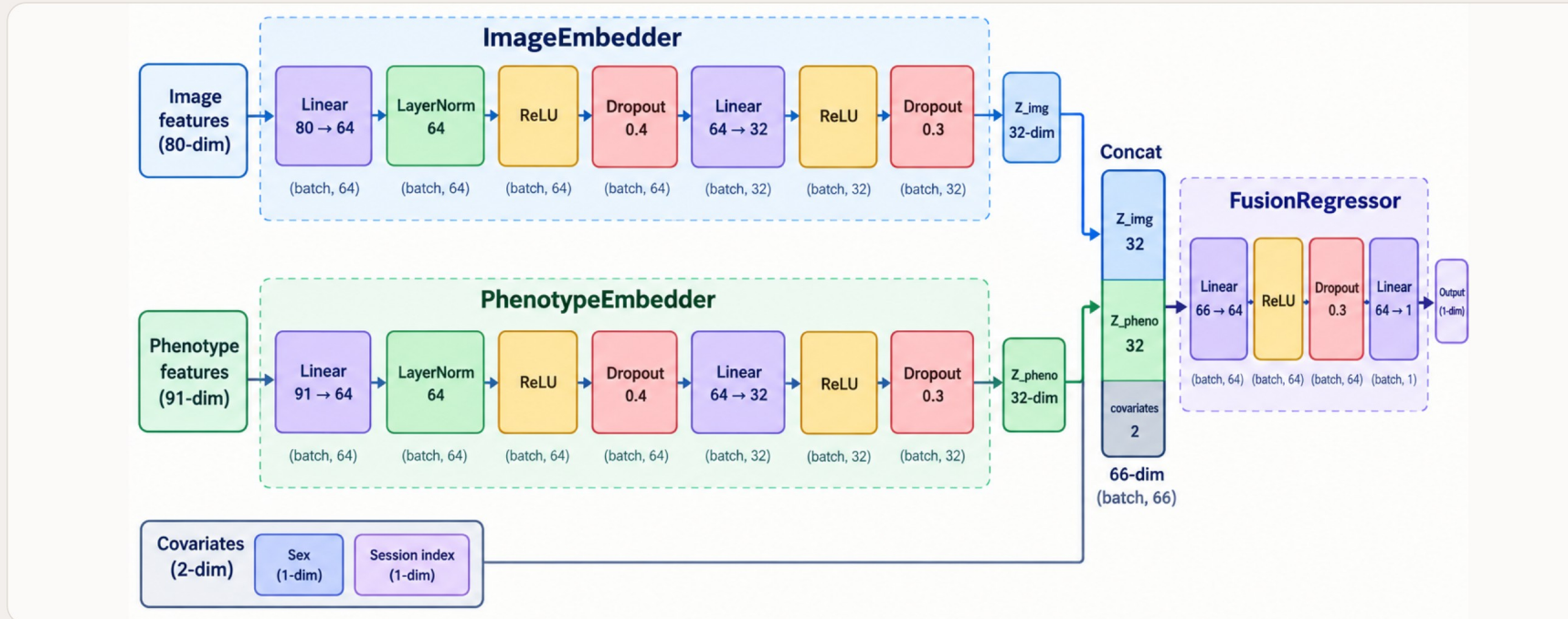
The five CNB task scores are standardised and averaged into the target — and held out of the inputs.

Every block is swappable

Genotype or omics enters as one more branch; nothing downstream changes.

THE MODEL

Two encoders, one head, random effects in the loss



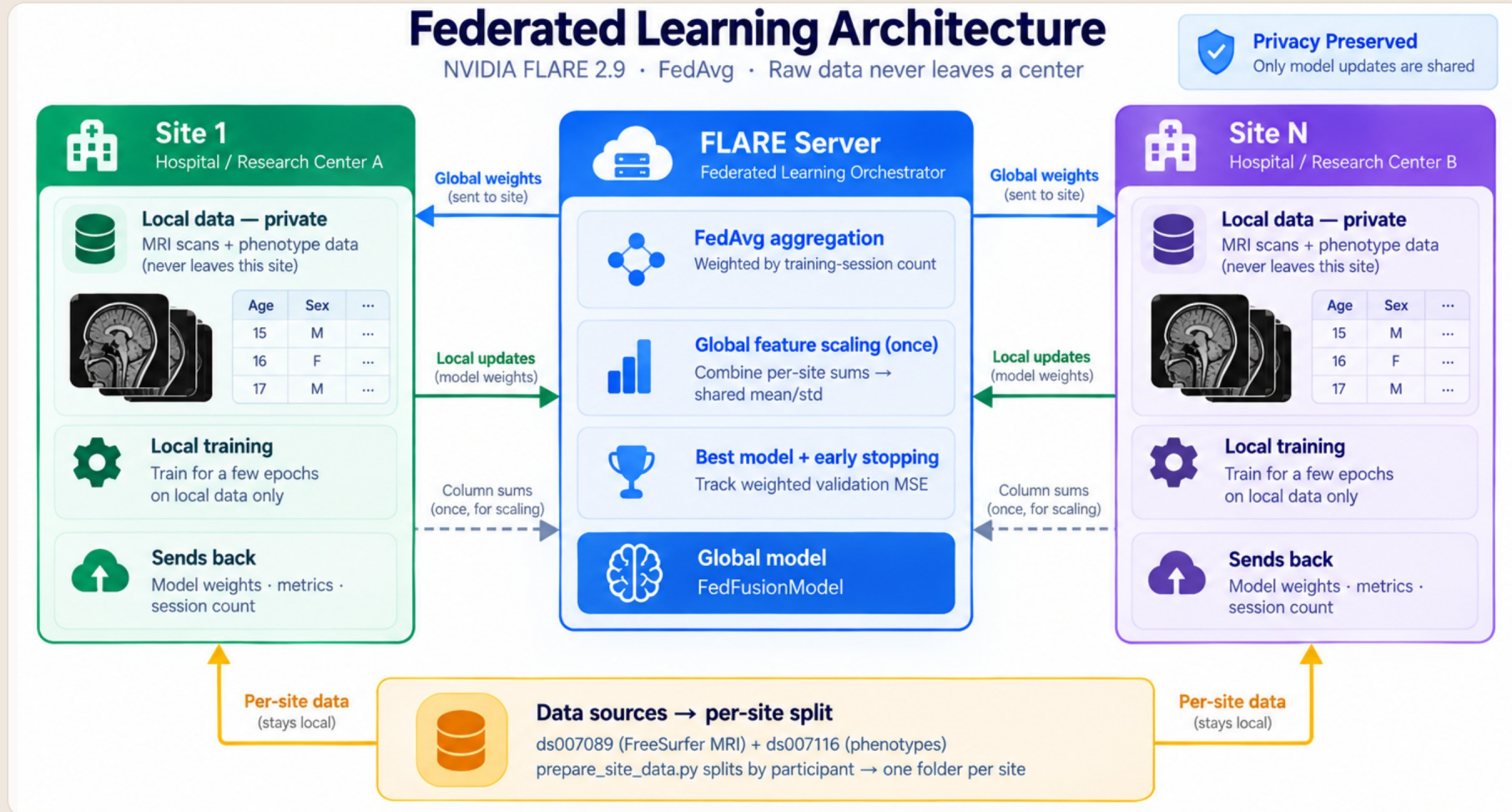
LMMNN loss — longitudinality, learned jointly

$$\text{loss} = \frac{1}{2} (r^T V^{-1} r + \log |V|), \quad V = \sigma^2_{\text{subject}} \cdot ZZ^T + \sigma^2_{\text{error}} \cdot I$$

Why it matters

A per-subject random intercept, optimised with the weights — no EM, no alternating steps. Subject-grouped minibatches; a subject's sessions never cross the split.

Nothing but aggregates leaves a site



The server sees weights, metrics, session counts and one set of aggregate column sums — **never a record**. Runs as simulator, POC processes, an exported job, or a provisioned multi-machine deployment with Docker.

RESULTS

FedAvg matches pooled-data performance

0.279 vs 0.275

Federated against pooled R^2 , five seeds each — a gap far inside the seed-to-seed spread, and 2.2× what one site reaches alone.

Setting	R^2
Offline FedAvg, 5 seeds	0.279 ± 0.039
Centralized pooled, 5 seeds	0.275 ± 0.043
Single site alone, 5 seeds	0.126 ± 0.038
NVFLARE simulator, 4 sites, seed 0	0.128

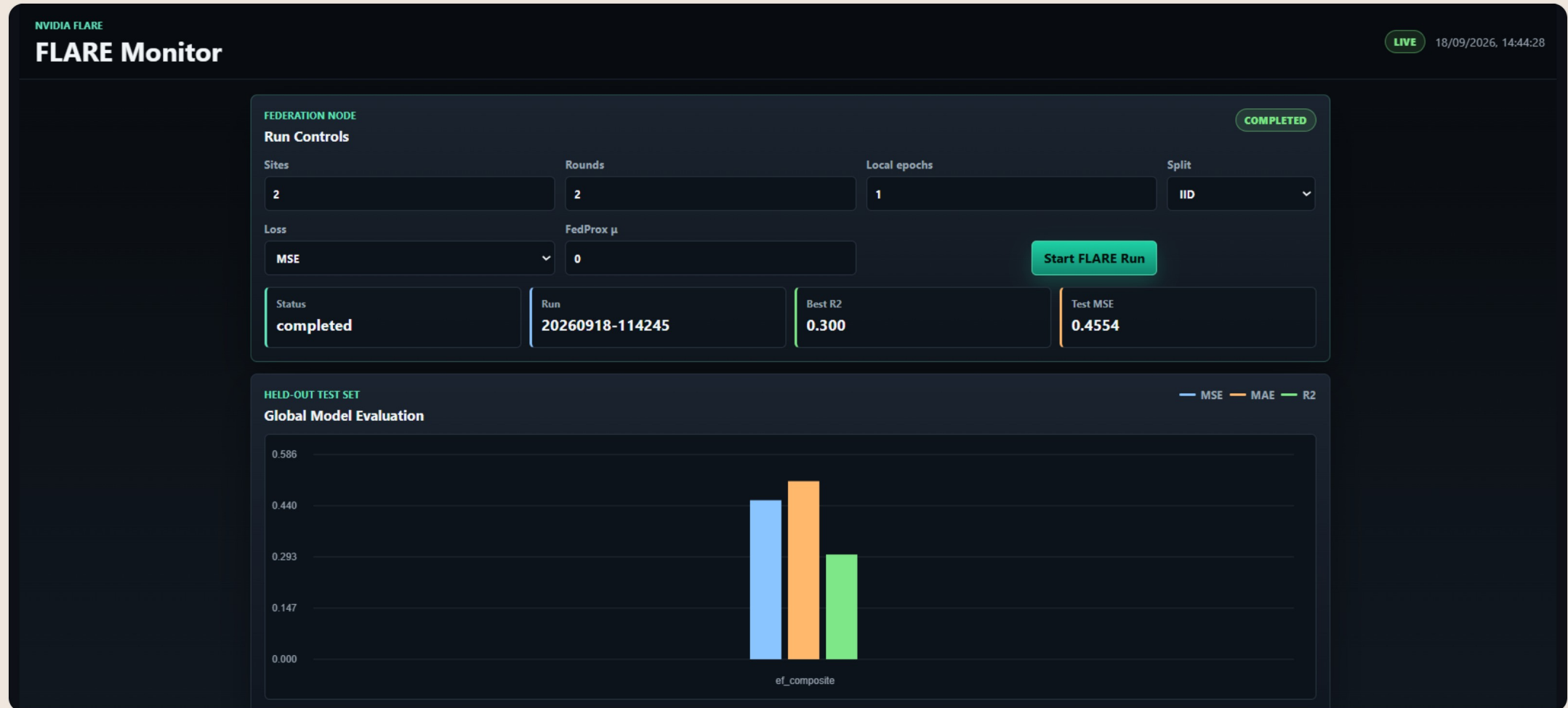
Centralized fusion model — train vs. validation MSE



Target: EF composite, same 39-session held-out test throughout, held out by participant. IID 4-site split, MSE loss. LMMNN sanity run, seed 0: $R^2 = 0.239$ against a mean-baseline MSE of 0.652.

IT RUNS

One dashboard drives the whole federation



Sites, rounds, local epochs, IID or age-based split, MSE or LMMNN, FedProx strength — set them, press start, and the global model is evaluated on the held-out test set. Shown: a 2-site, 2-round run, best R² 0.300. Dockerised, one **compose up**.

WHAT IT UNLOCKS

A blueprint a Nordic biobank can adopt

Modular

Swap the image embedder, the target (one composite or five sub-scores), or the loss — the rest of the pipeline adapts.

Multimodal

Late fusion is a clean seam: genotype, omics or actigraphy enter as one more embedder into the same head.

Cooperative

Global scaling from aggregates shows harmonisation is possible even when centres instrument differently.

Longitudinal

Session-level rows with a per-subject random effect today; a site or scanner effect is the next natural term.

Next

Real multi-machine deployment · per-site test sets · BLUP subject-specific correction at inference · a site/scanner random effect · a data-governance schema check. Code, docs and results are open on GitHub.